Better Sensor Prediction, Worse Flight: A Case Study in Quadrotor Adaptation

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Abstract: Can self-supervised sensor prediction tell us whether to accept an estimator update before flight? We study this question in simulation with onboard sensing, an ESKF–GRU estimator, and a frozen quadrotor controller. Adaptation on 120 seconds of unlabeled recordings improves independent sensor loss for all 30 primary heads, yet mean failure-penalized 30-second tracking RMSE rises from 0.224 to 0.266 m and success falls from 80.0% to 65.5%. Removing the head's offset parameters attenuates the regression. In a post-hoc IMU-bias diagnostic, removing optical-flow prediction from the adaptation objective reduces horizontal correction RMS by 76% and restores tracking to 0.132 m (frozen: 0.134 m; full adaptation: 0.194 m), while retaining nonzero vertical corrections. Twelve-second evaluation largely conceals the tracking penalty. These simulation-only results show why loss composition and evaluation horizon matter when using sensor consistency to assess estimator updates. Anonymous project page and simulation video: https://sensor-loss-flight.netlify.app.

Keywords: Transfer monitoring, self-supervised adaptation, quadrotor control

1 Introduction

Micro-quadrotors make sim-to-real transfer difficult: short flight endurance, limited onboard sensing, and costly repeated trials constrain adaptation and validation. These constraints motivate adapting an estimator from a short onboard recording, then testing its consequences in simulation before hardware deployment. But improved sensor consistency need not improve closed-loop control. We ask whether lower prediction loss on independent sensor recordings is sufficient evidence to accept an estimator update before flight.

We test this criterion in a StampFly-motivated simulation. Native-firmware recordings from physical StampFly hardware inform portions of its sensor-noise and timing statistics; remaining uncertainties and plant variation use explicit assumptions and controlled perturbations. A small velocity-correction head learns through future-sensor prediction; the estimator and RAPTOR controller [1] remain frozen. Every 120-second head improves held-out sensor loss across five primary conditions, yet tracking worsens on average (Fig. 1). The mismatch persists without coefficient clipping.

The classical error-state Kalman filter (ESKF) performs better overall than the frozen hybrid: 0.198 versus 0.224 m mean failure-penalized tracking RMSE. The acceptance question compares the same hybrid before and after adaptation, holding its backbone and controller fixed.

Learned filters [2, 3] and self-supervised inertial estimation [4, 5] motivate learning from sensor consistency. Objective mismatch has been studied for learned dynamics [6]; we examine it at the estimator–controller interface. Our results give a controlled counterexample to accepting updates on sensor loss alone. Ablations link the tracking penalty to learned velocity offsets and optical-flow prediction. Short evaluations also hide much of the later degradation. The study is simulation-only.

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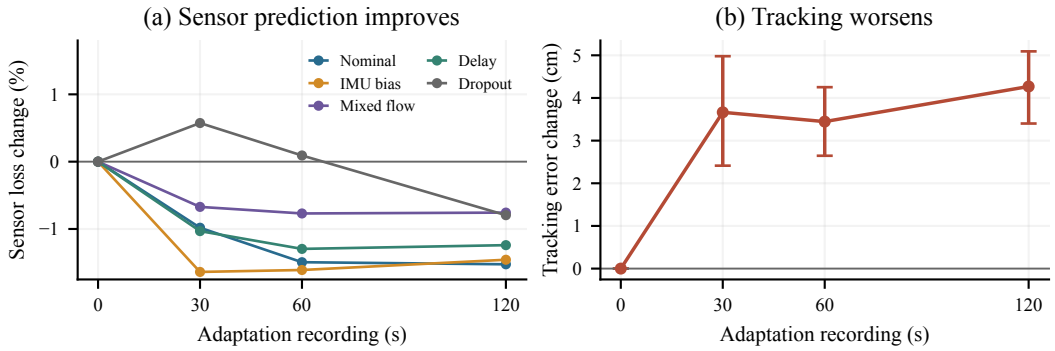


Figure 1: At 120 seconds of adaptation, independent sensor loss improves across all five primary conditions, while mean tracking error increases. Left: sensor-loss changes averaged over six heads per condition. Right: tracking changes relative to the frozen hybrid, with paired 95% flight-cluster intervals. Budgets share recordings and initialization; points are not independent trials.

2 Method

The testbed models the M5Stack StampFly (K138) [7], combining L2F dynamics [8] with native sensor geometry and a 15-error-state ESKF [9, 10]. We vary sensor bias, delay, dropout, and plant parameters while keeping the controller frozen and sensor timing causal. Flow depends on velocity divided by height; neither flow nor downward range provides an absolute horizontal position anchor. Published RAPTOR StampFly flights used external state measurements [1].

Measurements are fused in acquisition-time order and propagated to action time. A causal motor/body model predicts angular rate from delivered gyro and past commands. Physics and policy evaluation run at 400 Hz; policy recurrence and the GRU run at 100 Hz. The GRU corrects velocity, orientation, and angular rate. Truth supplies pretraining and evaluation only.

Only a zero-initialized head with 12 parameters, $W \in \mathbb{R}^{3\times3}$ and $b \in \mathbb{R}^3$, is adapted:

$$\delta v_k^b = 0.1 \tanh\left(W \bar{R}_k^\top \bar{v}_k + b\right), \tag{1}$$

$$\hat{v}_k = \bar{v}_k + \bar{R}_k \delta v_k^b + 2 I_k e_z \left(h_k - \hat{p}_{k-1,z}\right), \tag{2}$$

$$\hat{p}_k = \hat{p}_{k-1} + \Delta t\, \hat{v}_k, \qquad \Delta t = 0.0025 \text{ s}. \tag{3}$$

Here $\bar{v}_k$ and $\bar{R}_k$ are the frozen velocity and body-to-world rotation, h_k is tilt-corrected range, I_k indicates valid range, and e_z is the vertical unit vector. Relative position starts at zero. The head adds a body-frame correction bounded by 0.1 m/s per component; its coefficients are clipped to ±0.5. RAPTOR receives relative position, rotation, relative velocity, angular rate, and past motor commands through its unchanged 22-dimensional interface.

We estimate acceleration from the preceding 100 ms of corrected velocity. We predict accelerometer, flow, and range 50–150 ms ahead in acquisition time, assuming constant acceleration and body rate. The entire future flow interval is withheld. We fix sample eligibility using unadapted estimates and sensor validity. Scaled Smooth-L1 terms (weights 0.1/1/1) plus coefficient regularization train only the head, using 60 Adam steps at 10^{-2} for every nonzero recording budget.

3 Experimental Setup

We test five sensing conditions: nominal randomized sensing, post-boot IMU bias, mixed flow error, delay, and dropout. Each has three persistent simulated sensor shifts. For each shift, adaptation uses twelve independent 10-second ESKF-controlled clips, with a reset before each clip. Budgets of 30, 60, and 120 seconds use the first 3, 6, and 12 clips. We exclude two seconds of startup and reserve the final 20% for validation, separated from training by a 200-ms guard. Budgets count the full recording time.

Independent sensor recordings cover hover, ellipse, figure-eight, multisine, and excursion motion, weighted equally. We broadened the initial hover-only check by adding four motions, without refitting or selecting heads. Control evaluation uses 20 fresh flight seeds and 30-second trajectories. Sensor and plant draws are matched across methods. Seeds are assigned 7/7/6 across the three shifts. Two model optimization seeds share a pretrained initialization. Estimator and recurrent states reset before each flight. Known-rest boot calibration and initial relative heading are shared priors.

For intervals, we resample 20 paired flight-seed clusters, conditional on the three shifts and two models. The 30 primary heads ($5 \times 3 \times 2$) share recordings and initialization, so they are not independent trials.

Our tracking metric, E_{30}, is true position RMSE over 30 seconds. After early termination, we pad each missing squared error with $\max(1\,\mathrm{m}^2, e^2_{\mathrm{last}})$. This penalizes failure without implying observed motion. Success requires completing the flight with RMSE at most 0.25 m. Scenario partitions are disjoint, and final control results never select a checkpoint.

4 Results

4.1 Sensor prediction and tracking

Estimator	E_{30} (m)	Success (%)	Completion (%)
ESKF	0.198	80.0	90.0
Frozen hybrid	0.224	80.0	90.0
+ adaptation, 120 s	0.266	65.5	88.5

Table 1: Primary comparison across five sensing conditions at 30 seconds. E_{30} includes failure padding. ESKF has 100 episodes; each hybrid row has 200, over the same 20 flight-seed clusters.

Adaptation worsens tracking and success relative to the frozen hybrid (Table 1), despite improving sensor prediction (Fig. 1). The paired tracking increase is 4.27 cm (95% interval [3.40, 5.09]); both model seeds regress. All 30 primary heads improve independent combined sensor loss, yet 26 worsen mean paired tracking. Across condition means, sensor loss decreases only 0.8–1.5%. Accelerometer and flow loss improve for all heads, while range loss worsens for four: an aggregate gain need not improve every sensing channel or motion.

Decoder-only ESKF gives 0.202 m; non-neural calibration retains 80% success despite worse mean tracking. Different initial threshold margins limit this comparison. Alternative failure penalties preserve the main regression (Appendix A).

4.2 Evaluation horizon

The success-rate gap is nearly absent at 12 seconds (87.5% adapted versus 88.0% frozen). It reaches 14.5 percentage points at 30 seconds (65.5% versus 80.0%) on the same flights. Of 200 adapted flights, 23 exit the environment and 46 finish above the tracking threshold. Exits occur at the 1-m position limit; they do not necessarily mean crashes. True-state RAPTOR succeeds on 20/20 tasks at 0.076 m.

Mean velocity-bias norm rises from 0.0114 to 0.0142 m/s, and terminal position-estimation drift from 0.301 to 0.373 m. This is consistent with accumulating estimation error: a constant 0.01-m/s velocity error adds 0.30 m of position error over 30 seconds. Bias and drift are kinematically related, so this agreement does not uniquely identify the cause.

4.3 Diagnosing the tracking regression

We lower the learning rate, keeping 120 seconds of data and 60 updates fixed. Smaller rates (10^{-4}–3×10^{-3}) reduce but do not remove the tracking penalty. Sensor validation within the recordings

Head / learning rate	ΔL (%)	E_{30} (m)	Success (%)
Frozen hybrid	+0.00	0.263	78.0
Full, 3×10^{-3}	-0.59	0.294	66.0
Offset only	-0.11	0.294	67.0
Velocity dependent only	-0.46	0.268	78.0

Table 2: Offset diagnostic at the sensor-selected learning rate 3×10^{-3}; no head clips. ΔL is the independent sensor-loss change. The frozen baseline differs from Table 1 because the ablation uses a separate evaluation set with ten fresh flight-seed clusters.

selects 3×10^{-3} for offset-only ($W = 0$) and velocity-dependent-only ($b = 0$) heads. We freeze the heads before 800 comparisons on ten fresh flight-seed clusters (Table 2).

At this rate, the full head lowers independent sensor loss by 0.59%, but tracking worsens by 3.15 cm [2.09, 4.17] and success falls from 78% to 66%. No reduced-rate head clips, so clipping is not needed for the mismatch. The full learning-rate results appear in Appendix A.

Offset-only adaptation gives similar tracking to the full head. Removing b improves tracking by 2.65 cm [1.70, 3.55] relative to the full head. Velocity-bias norm falls by 0.0020 m/s and drift by 0.051 m. Success matches the frozen baseline at 78%, while sensor loss improves 0.46%. This points to learned offsets, although capacities differ (three versus nine parameters) and $b = 0$ can still give a nonzero mean correction. Tracking remains slightly worse than frozen (0.50 cm [0.04, 1.00]).

Optical-flow loss ablation. We next run a post-hoc check under IMU bias. Accelerometer/flow/range loss weights change from $(0.1, 1, 1)$ to $(0.1, 0, 1)$. The estimator still receives optical flow. All other original settings, including learning rate 10^{-2}, stay fixed.

On 40 matched 30-second flights per method, tracking improves from 0.194 to 0.132 m (frozen: 0.134 m), and success rises from 77.5% to 100%. The paired improvement is 6.23 cm (95% interval [4.30, 8.30]). All six heads improve the reduced held-out loss (mean per-head reduction 5.94%), while mean original combined loss increases 0.06%.

On independent sensor recordings, horizontal body-frame correction RMS falls from 1.20 to 0.29 cm/s. Vertical RMS stays similar (1.15 versus 1.12 cm/s). This check reuses the original flight seeds (Appendix A.6).

5 Discussion and Conclusion

The ablations implicate learned velocity offsets and loss composition in the tracking regression. Under IMU bias, removing flow loss suppresses horizontal corrections while retaining nonzero vertical adaptation and recovering frozen-baseline tracking. For approximately fixed orientation, acceleration differences and downward range weakly constrain horizontal offsets, whereas flow depends directly on velocity. The intervention also reduces the overall correction magnitude. The generator quantizes endpoint flow, whereas the forecast is continuous; the intervention does not distinguish quantization from other sensor-model errors.

The flow-loss result is specific to IMU bias. Approximate sensing, three sampled shifts per condition, shared model initialization, and ESKF-controlled recordings limit generalization. Adapted control can visit different states. Physical transfer and embedded execution remain untested.

An aggregate sensor score can hide which state errors an update introduces. Acceptance should therefore account for the downstream control task and its intended horizon: here a 12-second check largely misses the later cost. Bias-aware calibration and task-aware monitoring are next steps.

This study is ongoing. Future work will validate the observed mismatch on physical StampFly hardware, including closed-loop flight experiments under real sensing conditions.

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A Appendix

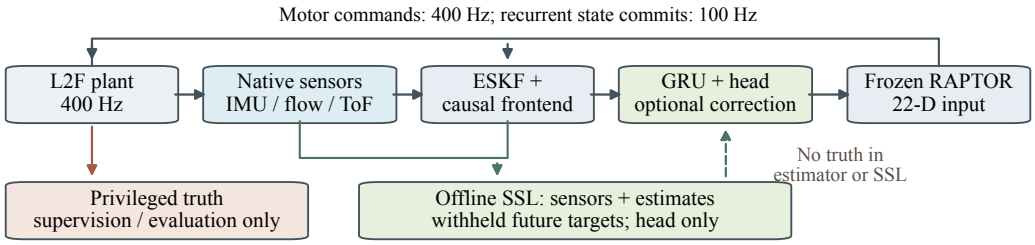


Figure 2: Causal control and separate offline adaptation. Truth generates simulated readings and supplies supervised pretraining and evaluation. Estimator inference and self-supervision consume delivered sensors, estimator outputs, and past commands; the policy weights remain frozen.

A.1 Implementation and sensing

The simulated IMU, flow, and range run at approximately 400/100/30 Hz. A 48-unit GRU uses 47 causal sensor/frontend and ESKF features. An encoder pretrained on 2,000 simulated flights receives 80 updates on 120 training and 20 validation clips of up to eight seconds. The two optimization seeds share initialization. The sensor-loss scales are 1 m/s^2, 1 count, and 0.05 m for accelerometer, flow, and range. Coefficient regularization is $0.01(\mathrm{mean}(W^2) + \mathrm{mean}(b^2))$; gradient norm is clipped to one. A zero-second budget preserves the initial head exactly. Original maximum holdout correction is 75.3% of the output bound, below the 95% saturation threshold.

Let R map body FLU to world FLU and $S = \mathrm{diag}(1, -1, -1)$ convert to sensor FRD coordinates. Specific force is $SR^\top(a - g_w)$, where $g_w = (0, 0, -9.80665)$. Flow uses $v^b = SR^\top v$, $\omega^b = S\omega$, and height $h = p_z$:

$$c = \frac{\Delta t_{\mathrm{flow}}}{0.00222} \begin{pmatrix} v_x^b/h + \omega_y^b \\ v_y^b/h - \omega_x^b \end{pmatrix}, \qquad r = \frac{h}{R_{zz}}. \tag{4}$$

The generator quantizes acquisition-endpoint motion scaled by interval duration; it does not integrate exact motion over the flow interval. The forecast uses the corresponding continuous approximation.

Added stresses are accelerometer/gyro bias ± 0.05 m/s^2/± 0.01 rad/s, flow scale 1.08 and bias up to 0.35 counts, IMU/flow/range delays 3–10/8–20/15–40 ms, and dropout rates 3/15/10%. Mass and inertia vary $\pm 10\%$ and motor time constants $\pm 20\%$. These assumptions extend beyond the partial log-derived statistics. Maximum desired altitude is 0.9422 m, leaving limited overshoot margin within the 1-m box.

A.2 Sensor-loss detail

Table 3 averages per-head relative loss changes over three shifts and two model seeds after equal averaging of five motion types. Set ranges first average model seeds within each shift. Combined percentages differ from component percentages because the combined score sums weighted absolute component losses.

A.3 Quantized-flow example

At $h = 0.5$ m, $\Delta t_{\mathrm{flow}} = 10$ ms, zero rotation, and forward speed 0.05 m/s, the noiseless native count function emits zero. Its continuous prediction is 0.45045 counts. Predicting 0.04 m/s instead reduces the implemented two-axis flow loss from 0.05073 to 0.03246 (36%) while adding 0.01 m/s velocity error. This deterministic calculation illustrates surrogate-model mismatch; it is neither a flight result nor a measured hardware effect, and does not quantify its causal share of the regression.

Domain	Accel (%)	Flow (%)	Range (%)	Combined: set range (%)	ΔE_{30} (cm)
Nominal	-1.45	-1.24	-4.16	[-2.83, -0.54]	+1.07
IMU bias	-1.60	-0.97	-8.19	[-3.21, -0.39]	+6.01
Mixed flow	-3.38	-0.60	-2.72	[-1.09, -0.53]	+1.37
Delay	-1.71	-0.89	-7.97	[-2.04, -0.79]	+4.56
Dropout	-1.64	-0.74	-0.38	[-1.04, -0.37]	+8.35

Table 3: Original 120-second adaptation. Lower is better. Set ranges summarize three adaptation sets, not confidence intervals.

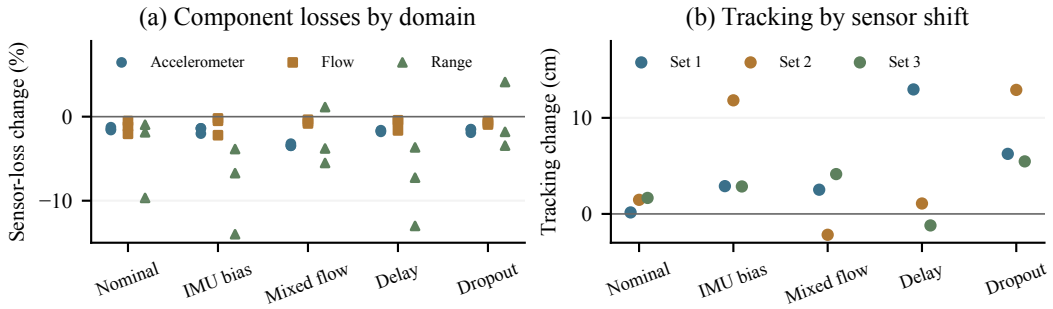


Figure 3: Original 120-second adaptation relative to the frozen hybrid. Left: component loss changes; points average two model seeds within each adaptation set. Right: tracking changes by condition and set. Shared shifts make these descriptive points, not independent population samples.

A.4 Additional results

The full original study contains 1,780 episodes, including separate flow-noise/systematic-flow controls and 240 transfers back to nominal sensing. For the primary 120-second comparison, last-error hold and a 0.5-m padding floor both give a 4.06-cm paired increase [3.26, 4.83]; a 1.5-m floor gives 4.89 cm [3.73, 6.18]. Only the near-tied 30-/60-second adaptation ordering swaps with the largest floor.

A separate systematic-flow condition favors the unadapted hybrid over ESKF (0.311 versus 0.375 m), but does not reverse the overall ordering.

Estimator	E_{30} (m)	Success, 12 s	Success, 30 s	Complete, 30 s
ESKF	0.198	88.0	80.0	90.0
Frozen hybrid	0.224	88.0	80.0	90.0
+ adaptation, 30 s	0.260	87.0	71.5	89.0
+ adaptation, 60 s	0.258	88.0	70.5	89.5
+ adaptation, 120 s	0.266	87.5	65.5	88.5
Kinematic ESKF	0.202	88.0	80.0	90.0
Calibration only, 120 s	0.235	89.0	80.0	90.0

Table 4: Primary five-condition study. E_{30} includes failure padding; success and completion are percentages. The 12-second scores use prefixes of the same 30-second flights. Classical/neural rows have 100/200 episodes over the same 20 flight-seed clusters. Reference rows isolate the filter, position decoder, and non-neural calibration.

A.5 Reproducibility

Frozen sources, head hashes, disjoint partitions, and causality, reset, and streaming/batch tests are retained. Original evidence was audited from 1,780 trajectories; follow-up records are frozen separately. The initial hover-only holdouts and report-only coverage supplement remain available. No final tracking outcome selects a checkpoint.

A.6 Diagnostic ablation details

Table 5 retains every tested learning rate. The smallest rate gives a 0.51-cm tracking increase [0.10, 0.91] with unchanged success. All original heads clip at least one coefficient; none of the reduced-rate heads does. Coefficient bounds and output saturation are distinct.

Head / learning rate	ΔL (%)	E_{30} (m)	Success (%)
Frozen hybrid	+0.00	0.263	78.0
Full, 10^{-2}	-1.15	0.309	62.0
Full, 10^{-4}	-0.05	0.268	78.0
Full, 10^{-3}	-0.31	0.284	69.0
Full, 3×10^{-3}	-0.59	0.294	66.0
Offset only	-0.11	0.294	67.0
Velocity dependent only	-0.46	0.268	78.0

Table 5: Full learning-rate and parameterization diagnostic; the evaluation set and loss definition match Table 2.

The flow-loss diagnostic keeps the original twelve 10-second recordings per shift, zero initialization, learning rate 10^{-2}, 60 Adam steps, regularization, clipping, backbone, and controller. Only the flow-loss weight changes. All six full-loss fits reproduce the saved head parameters exactly. No-flow heads freeze before 40 new evaluations on the original 20 flight seeds and three shifts (7/7/6), with two model seeds. Frozen/full-adaptation results are reused. This is a post-hoc, condition-specific diagnostic, not an independently validated adaptation method.

Estimator	L	$L_{\neg f}$	E_{30} (m)	Success (%)
Frozen hybrid	0.96785	0.07591	0.134	100.0
Original adaptation	0.95761	0.07160	0.194	77.5
No flow loss	0.96843	0.07136	0.132	100.0

Table 6: IMU-bias diagnostic: 40 paired 30-second episodes per row. L is the original sensor loss; $L_{\neg f} = 0.1L_{\text{accel}} + L_{\text{range}}$. Independent scores average five motions per head, then six heads, and omit coefficient regularization.

The reduced-loss percentage averages per-head relative changes; the 0.06% original-loss increase compares absolute means in Table 6. No-flow tracking differs from frozen by -0.23 cm (95% interval [-1.11, 0.59]), so superiority to frozen is unresolved.

Correction statistics pool the same independent five-motion holdouts after two seconds of startup. They measure the bounded body-frame head output before height feedback. After decoding with height feedback, RMS velocity change relative to frozen falls from 1.26 to 0.44 cm/s. The intervention therefore changes both loss composition and effective correction magnitude; their contributions are not isolated by this experiment.